a single *s*. When the components are multiplied together, they must equal the original numerator and denominator. In other words, the equation is placed into the form:

EQUATION 32-3
The factored s-domain. This form allows the s-domain to be expressed as poles and zeros.

$$H(s) = \frac{(s - z_1)(s - z_2)(s - z_3)\cdots}{(s - p_1)(s - p_2)(s - p_3)\cdots}$$

The roots of the numerator, $z_1$, $z_2$, $z_3$ ···, are the **zeros** of the equation, while the roots of the denominator, $p_1$, $p_2$, $p_3$ ···, are the **poles**. These are the same zeros and poles we encountered earlier in this chapter, and we will discuss how they are used in the next section.

Factoring an s-domain expression is straightforward if the numerator and denominator are *second-order polynomials*, or less. In other words, we can easily handle the terms: $s$ and $s^2$, but not: $s^3$, $s^4$, $s^5$, ···. This is because the roots of a second-order polynomial, $ax^2 + bx + c$, can be found by using the quadratic equation: $x = -b \pm \sqrt{b^2 - 4ac} \;/\; 2a$. With this method, the transfer function of the example notch filter is factored into:

$$H(s) = \frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$$

where:

$$z_1 = j/\sqrt{LC} \qquad p_1 = \frac{-R + \sqrt{R^2 - 4L/C}}{2L}$$

$$z_2 = -j/\sqrt{LC} \qquad p_2 = \frac{-R - \sqrt{R^2 - 4L/C}}{2L}$$

As in this example, a second-order system has a maximum of two zeros and two poles. The number of poles in a system is equal to the number of independent energy storing components. For instance, inductors and capacitors store energy, while resistors do not. The number of zeros will be equal to, or less than, the number of poles.

Polynomials greater than second order cannot generally be factored using algebra, requiring more complicated numerical methods. As an alternative, circuits can be constructed as a *cascade* of *second-order* stages. A good example is the family of analog filters presented in Chapter 3. For instance, an eight pole filter is designed by cascading four stages of two poles each. The important point is that this multistage approach is used to overcome limitations in the *mathematics*, not limitations in the *electronics*.